Module 3 Basic Content:

Water properties (Add buoyancy and settling to this)

- Universal solvent: water can dissolve many substances and hold them in a liquid form (solution); e.g., salt solution, sugar solution, bubble solution, Kool-Aid
    - Solubility: temperature affects water's ability to dissolve things. The warmer the water, the more it can dissolve. This is why tea that is sweetened before it cools is often much sweeter than tea sweetened after it has cooled or had ice added to it.
        - Odd factoid: a super-saturated solution is formed by heating water and dissolving as much of a solute (e.g., salt, sugar) into it as possible, then *slowly* cooling the solution – if it is cooled too fast, part of the dissolved solute will reform into a solid. The slowly cooled solution is called super-saturated because it holds much more of the dissolved solute than it should at that temperature. If super-saturated enough, simply bumping the container may cause the solute to reform into a solid in the solution. Super-saturated solutions are used to make rock candy.
    - Odd factoid: water in nature normally has several things dissolved in it such as oxygen and minerals; often the compounds dissolved in water provide it with additional properties such as alkalinity and the ability to transmit electricity
- When in small quantities and in small spaces, water adheres (sticks) to itself and other things
    - Surface tension: water molecules on the outer surface adhere to each other more forming a slightly denser outer layer; this is why water forms drops and beads up in small amounts – larger water beads lose their spherical shape and become flatter due to gravity
        - Odd factoid: many organisms have evolved to take advantage of water's surface tension, such as water striders, giant water bugs, fisher spiders, diving bell spiders, green basilisk lizards, waterfowl (ducks and geese), tail-walking dolphins, and even humans.

    - Capillary action: in tubes, water will “climb” up the sides of the tube. The smaller the tube the higher the water can climb.
        - Meniscus: capillary action found in larger-diameter tubes; due to the larger diameter, the water can only climb a short distance, producing a stretched out “U” shape when viewed from the side.
        - Odd factoid: capillary action is how water moves from the roots of plants up to the leaves; for the water to move extensively in plants using capillary action, water vapor must transpire out of the leaves at the top to overcome gravity and increase the upward movement.

- Density: the amount of “stuff” in a given space. In general, water’s density is given as 1 gram per milliliter, which is about the amount of water needed to fill the cap of a soda bottle. Even though this is a standard measurement, water’s density actually varies based on both what is dissolved in it and its temperature.
    - Saltwater is more dense than freshwater due to the dissolved salt. When a salt dissolves in water it separates into smaller ions that float among the water molecules (more stuff in the same space). Normally when freshwater streams empty into bodies of saltwater, the two waters mix forming “brackish” water but when the waters are colder and more dense, this mixing does not happen as well. There are many bodies of water which have areas of increased salinity at the bottom of them due to the saltier water sinking to the bottom.
        - Odd factoid: the Dead Sea is one of the saltiest bodies of water on Earth at 34% salinity, which also makes it one of the densest bodies of water. Due to the Dead Sea’s high density, people find it difficult to swim in the water because they float much higher than normal.
    - As water cools and the molecules move closer together, the water becomes more dense (same amount of stuff, smaller space). Upon freezing, the molecules form a crystal latticework which forces the water molecules farther apart making the ice less dense than liquid water (same amount of stuff, larger space). This is why ice cubes and ice bergs both float in water – no matter how huge the chunk of ice is, it will always float in water.

    - Frost wedging: water increases in volume when it freezes so it takes up more space. This explains why you see a bump on the top of fresh ice cubes, why water pipes break in the winter when frozen (even metal ones), and why a can of Dr. Pepper is likely to split open if forgotten in the freezer. Frost wedging is what eventually reduces mountains to piles of rubble as it is a form of erosion alongside wind and rain. Frost wedging is the reason why so many potholes develop on roads in the winter because when water seeps into cracks in the road and freezes, the ice pushes the sides of the crack further apart, allowing more water in, speeding the process.

- Heat capacity: water can hold a lot of heat, so it takes a long time to warm it up and cool it down; there's a reason for the saying "a watched pot never boils". Due to this property, the presence of water in a natural system helps moderate (slow down) temperature changes in the environment. Deserts are well known for rapid temperature swings between day and night because there is almost no humidity (water vapor in the air) to slow down the heating and cooling of the air. The opposite of this is the reason the climate tends to be milder near coastal areas – because the large body of water and high humidity levels slow down the heating and cooling of the air above the water. (Interestingly, since the land next to a large body of water cools and heats faster than the water, this is what causes the sea breeze to kick up at sunup and sunset, due to the temperature difference.) Water's high heat capacity is the reason why the main temperature control system for nuclear power plants is cooling towers filled with water – and why many of these power plants are found on a river or large body of water.

    - Odd factoid: the heat capacity of water is important to the organisms that live in it. All organisms have a "range of tolerance" for many things in their environment, and temperature is one of them. Organisms that live on land or live in large bodies of water have the ability to find shelter or move from one place to another to deal with temperature extremes, but organisms that live in smaller bodies of water do not have this option. This is why water's heat capacity is so important, because this is what keeps the temperature of a body of water (at least part of it) always within the organism's range of tolerance. During the heat of summer, the depths of a pond will generally remain cool enough for most fish to remain comfortable (this is why fishing is best at night during the summer) and during the cold of winter, those same deep waters will remain warm enough. The exception is if the pond freezes over; when that happens, the fish are more likely to be found in

the shallows because the light transmitted through the ice tends to warm those waters by a couple of degrees.

    - Odd factoid: human body temperature is also managed using the water in several ways. When we are hot the blood vessels in our skin dilate (expand) to move more blood through faster, so excess heat can be released through the skin (the expanded vessels make our skin looks pinker); when we are cold we look paler because the blood vessels in the skin contract to keep most of the blood flow in the body's core (head, chest, and abdomen) to conserve heat. Another thing humans do when they are hot is sweat. As water in the form of sweat is released from the pores in our skin and evaporates, heat is removed from our bodies as it is used to make the water evaporate. A third way in which humans cool themselves is through increased respiration (breathing or panting). Like sweating, body heat is used to vaporize water in our respiratory tract and with each breath we exhale moist air and heat into the atmosphere.

- Alkalinity: pure water has a pH of 7, but most water found in nature ranges from a pH of 6.8 to 9 (lower than 7 is acidic, higher than 7 is alkaline/basic). Due to having minerals dissolved in it (e.g., calcium carbonate), most fresh water is alkaline (pH above 7) – and the organisms that live there are happy about it.

    - Odd factoid: bottled water companies that purify water prior to bottling often add minerals back into the water so that it will taste like…water

- Electrical conductance: Pure water is an excellent insulator that will prevent electricity from being conducted through it, but most water is not pure. Water with dissolved salts or other minerals in it will transmit electricity, so almost all water found in nature will transmit electricity. The dissolved salts and minerals provide loose ions which give electrons (electricity) a pathway to travel through the water.

    - Odd factoid: While standing under a tree in a lightning storm will ensure you are not the tallest thing around to draw a lightning strike, you will almost certainly be struck anyway as the bolt of lightning travels down the tree into the ground and is transmitted through the water in the soil to anything or anyone standing on it.